ENGINEERING: THE DESIGN PROCESS, CAD, AND 3D PRINTING

Course Details:
Meeting Time and Place: 7th Period in Room C
Instructor: Matthew Bardoe

- **Email**: mbardoe@wellington.org
- **Office Number**: A206
- **Availability**: Task Time is a great time, except on Wednesdays where I often have an MS Dive. But look for me at other times as well. I have 1st and 3rd period free all year.

Course Overview:

PREREQUISITE: Integrated Science 1 or Department Approval
COURSE DURATION: 1 Trimester
COURSE CREDIT: 1/3 Physical Science Credit (Digital Literacy Badge)

Materials:

Technology: You will need a laptop for this class. A mouse is also strongly recommended; you may bring your own or borrow one while in the classroom.

Course Expectations:

This course introduces students to foundational concepts in engineering through hands-on exploration, CAD modeling, and digital fabrication. Students will engage in the engineering design process as they explore real-world problems, learn industry-standard software (OnShape), and create functioning prototypes using 3D printing and other Maker Space tools. Emphasis will be placed on creativity, iteration, communication, and problem-solving.

By the end of the course, students will:

- Create accurate, fully constrained sketches and use them to build three-dimensional parts.
- Plan and construct a CAD model from measurements or a dimensioned drawing.
- Create engineering drawings that communicate enough information to reproduce a part.
- Prepare a CAD model for FDM 3D printing and account for manufacturing constraints.

• Select tolerances and clearances based on physical testing rather than nominal dimensions alone.

• Construct assemblies and evaluate how parts fit, move, and interact. • Build robust parametric models that can adapt to changing requirements.

• Use named versions and branches to preserve milestones, explore alternatives, and explain design decisions.

• Apply an iterative design process: define, ideate, model, fabricate, test, revise, and communicate.

Attendance: Being in class is important, there is no substitute, but when you miss please speak with me about what you have missed and whether we should meet outside of class to cover any essential ideas.

Preparation: Get good amounts of sleep after each class. Sleep is when your brain organizes and stores all that you learned that day. If you don't get enough sleep all the learning you have done gets lost.

Assessments & Grading:

Homework: 30%
Projects: 70%

Late Homework:

Late work will generally receive full credit. However, if a student has three or more overdue assignments at the same time, those assignments may receive a 10% deduction. Homework will be accepted until the later of one week after its due date or the final assessment for that unit. After that point, the assignment may no longer be submitted for credit.

Late Projects:

Projects should be submitted on time. If you anticipate missing a deadline, contact me beforehand to propose a reasonable extension and provide a brief explanation—even if it feels lame. Projects submitted without prior approval or after an agreed-upon extension will receive a 10% deduction.

Faculty Commitments:

- Providing timely and meaningful feedback
- Maintaining an up-to-date gradebook
- Communicating academic concerns early to you, your advisor, and parents
- Creating a classroom where questions and mistakes are part of learning
- Being available to support your growth through office hours and extra help opportunities

Topics Overview:

Unit	Focus	Days
1	CAD Foundations and Precise Parts	10
2	Design for 3D Printing	8
3	Assemblies and Interacting Parts	3
4	Parametric Design and Version Management	8

* Note: The schedule is flexible and may adapt to the class pace.

Academic Integrity:

AI Usage: AI Acceptable Use Rating Scale

All Projects and homework assignments for this course can utilize up to Level 2 (unless noted explicitly)

Level 0: No Use
- AI and Generative AI are not permitted for use in any capacity.

Level 1: Limited Use
- AI can be used for basic assistance, such as spell-checking or grammar suggestions.
- Generative AI use is restricted to idea generation, with all final work being original.

Level 2: Moderate Use
- AI tools can assist in research and data analysis, but users must critically evaluate and verify the information.
- Generative AI can be used for draft creation, but substantial editing and personalization are required.

With express permission you may use:

Level 3: Enhanced Use
- AI can be integrated into project workflows, providing advanced insights and facilitating complex tasks.
- Generative AI can contribute significantly to creative processes, with proper attribution and reflection on its role.

Level 4: Full Use
- Full integration of AI in learning and teaching processes is encouraged, with an emphasis on ethical considerations and transparency.
- Generative AI is used extensively, with a clear understanding of its capabilities and limitations

Plagiarism Policy

From the student handbook: Plagiarism is a category of cheating and is the act of misrepresenting ideas or writing. Plagiarism includes representing others' ideas and writing as your own; self-plagiarism is representing formerly conceived ideas and previously written materials as new work and also includes submitting work that was generated by artificial intelligence (AI) programs. All cheating and plagiarism must be reported by faculty members to the Academic Dean, who will determine an appropriate response.

Cheating Policy

Cheating shall be defined as gaining or attempting to gain unfair academic advantage over another student through dishonest means. If the episode involves two or more students knowingly exchanging information, then all students involved are guilty of cheating. It is therefore important for each student to protect their own work and to understand any parameters for sharing it.